SOFTWARE ENGINEERING PROJECT

ACADEMIC CALENDAR

GROUP

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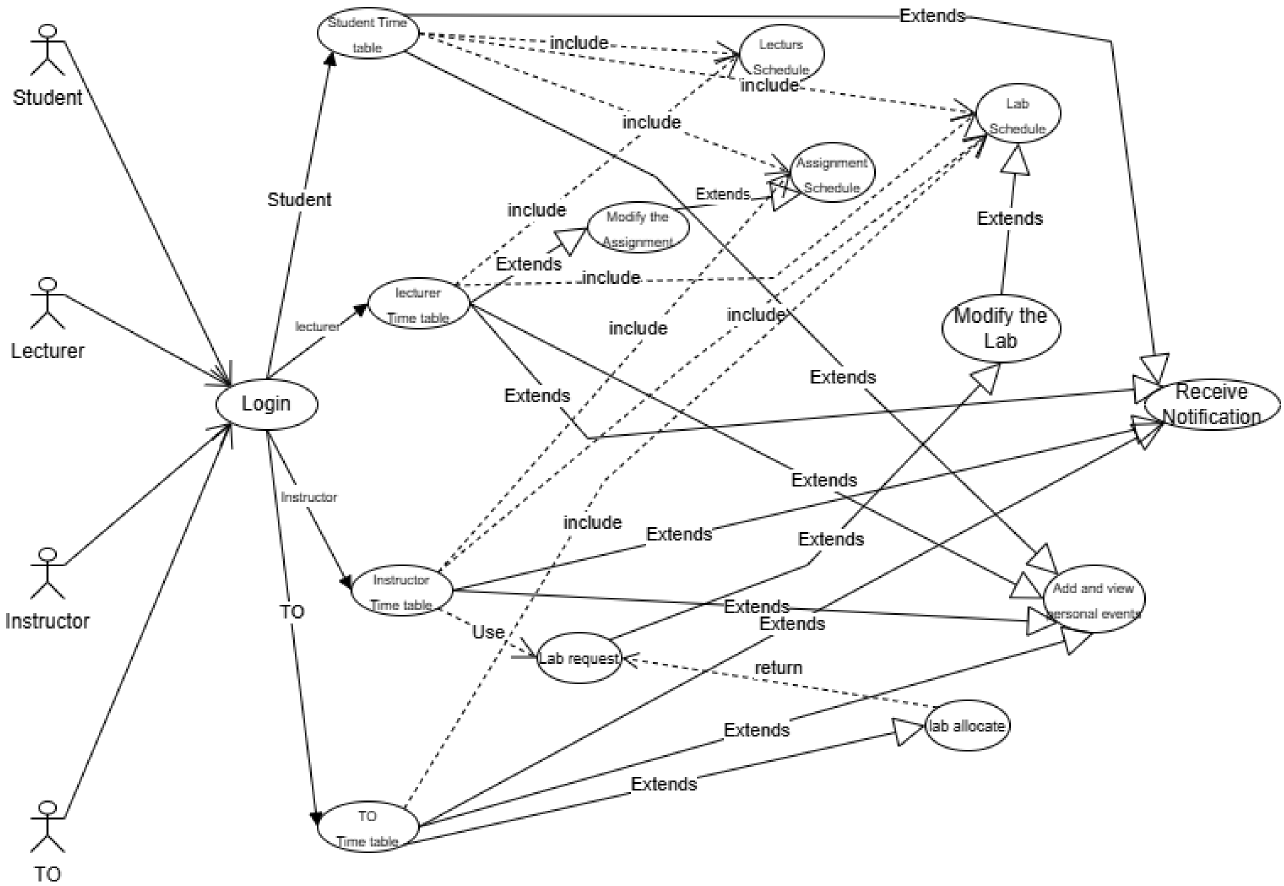
USERASE DIAGRAM

The Use Case Diagram for the **Shared Academic Calendar Mobile Application** illustrates the functional interactions between the system and its primary users through a role-based access model. The diagram defines how different actors interact with the system to manage and view academic schedules, events, and laboratory activities.

The system consists of four main actors: **Student, Lecturer, Instructor, and Technical Officer (TO)**.

- **Students** interact with the system to view lecture schedules, laboratory schedules, assignments, and event statuses such as upcoming, finished, postponed, or cancelled. Students are also allowed to create and manage **personal events**, which are private and visible only to the individual student. In addition, students receive notifications related to academic activities.
- **Lecturers** interact with the system to manage academic content related to their own subjects only. They can create lecture schedules, assign assignments, post special academic events, and publish announcements. Lecturers can also view laboratory schedules relevant to their subjects and monitor their status (upcoming, finished, postponed, or cancelled).
- **Instructors** are responsible for laboratory-related scheduling. They can schedule laboratory sessions for students, view assigned lab schedules, update lab status, and send lab scheduling requests to the Technical Officer for resource allocation.
- **Technical Officers (TOs)** handle laboratory resource management. They receive lab requests from instructors, verify equipment availability, allocate laboratories, and monitor all laboratory schedules along with their current status (upcoming, postponed, or cancelled).



NON-FUNCTIONAL REQUIREMENTS

1) Usability (Product Requirement)

The system shall provide an intuitive and easy-to-use interface that follows native mobile UI conventions.

Questions:

1. Can a new user navigate the app and view their schedule without prior training?
   - The application provides an intuitive interface with clear labels and a default “Today’s Schedule” view on launch, allowing new users to access their schedules without any prior training.’

2. Is the navigation consistent across lecture, lab, and event screens?
   - The application follows a consistent navigation structure and layout across all screens, ensuring uniform user interaction for lectures, labs, and events.

2) **Performance (Operational Requirement)**
The system shall synchronize schedules and notifications with minimal delay.
Questions:

1. Are newly added or updated events reflected in the application without noticeable lag?
   - Yes. Schedule updates and changes are synchronized promptly, ensuring that newly added or modified events are reflected in the application with minimal delay.

2. Does the "Today's Schedule" view load within an acceptable response time under normal network conditions?
   - Under normal network conditions, the "Today's Schedule" view loads within an acceptable response time, providing a smooth user experience.

3) **Security (Product Requirement)**
The system shall protect user data and restrict access based on user roles.
Questions:

1. Can students access only their own personal events and academic schedules?
   - Role-based access control ensures that students can view only their own personal events and assigned academic schedules, preventing access to other users' data.

2. Are unauthorized users prevented from modifying schedules or assignments?
   - The system enforces authentication and authorization mechanisms that prevent unauthorized users from creating, updating, or deleting schedules and assignments.

4) **Reliability (Product Requirement)**

The system shall operate reliably and maintain correct schedule data without loss or inconsistency.

1. Does the system retain all schedules correctly after app restarts or crashes?
   - The system securely stores schedule data in the backend database and uses local caching mechanisms, ensuring that all schedules are retained correctly even after application restarts or unexpected crashes.
2. Are events and lab schedules consistently displayed without missing or duplicated entries?
   - The system enforces structured data management and validation rules, which ensure that events and lab schedules are displayed consistently without missing or duplicate entries.

5) Availability (Operational Requirement)

The system shall be available for users during normal academic hours with minimal downtime.

1. Is the system accessible whenever students or staff attempt to view schedules?
   - The system is designed to be accessible during normal academic hours, allowing students and staff to view schedules whenever required.

2. Does the system recover quickly after temporary service interruptions?
   - The system supports rapid recovery after temporary service interruptions, restoring normal access without significant impact on user activity.